

# LINEAR AND ROBUST CONTROL SYSTEMS

## ELEC 9731



### 10 - MIMO Poles and Zeros

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**Slide 1**

## **Topics**

- 1** State Space View
- 2** Transfer Function View
- 3** Matrix Fraction Description View

## **References**

Kailath chapter 6

# WHAT IS A POLE?

- Transfer function view (SISO)

$s_p$  is a pole of  $P(s)$  if  $P(s_p) = \infty$

with  $P(s) = \frac{b(s)}{a(s)}$  ,  $a, b$  coprime polynomials

$\equiv$  to  $s_p$  is a root of  $a(s)$ .

- Dynamic or SS view (SISO) .

Consider  $\dot{x} = Ax + bu$  with no forcing. We look for natural modes i.e. response of the form  $e^{s_p t} x_0$  with no forcing. So set  $u = 0$ .

$$\begin{aligned}\Rightarrow 0 &= \dot{x} - Ax \\ &= s_p e^{s_p t} x_0 - A e^{s_p t} x_0 \\ &= (s_p x_0 - A x_0) e^{s_p t}\end{aligned}$$

This is possible iff  $Ax_0 = s_p x_0$  i.e.  $s_p$  is an eigenvalue of  $A$ .

i.e.  $|s_p I - A| = 0 = a(s_p)$

$\Rightarrow x_0$  is a right eigenvector.

- So SS & TF view agree that  $a(s_p) = 0$  but SS adds notion of an associated direction.

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## WHAT IS A ZERO? (SISO)

A zero is a 'frequency' yielding null output with nonnull input.

Suppose  $u = u_0 e^{s_o t}$ . Look for response of the form  $x = x_0 e^{s_o t}$  yet  $y = 0$

$$\begin{aligned}\Rightarrow 0 &= \dot{x} - Ax - bu \\ &= s_0 x_0 e^{s_o t} - Ax_0 e^{s_o t} - bu_0 e^{s_o t} \\ \Rightarrow 0 &= s_0 x_0 - Ax_0 - bu_0\end{aligned}$$

Also

$$\begin{aligned}\Rightarrow 0 &= y = (c^T x_0 + du_0) e^{s_o t} \\ \Rightarrow 0 &= c^T x_0 + du_0\end{aligned}$$

## ZEROES (SISO)

Collect these together giving

$$\begin{pmatrix} s_0 I - A & -b \\ -c^T & -d \end{pmatrix} \begin{pmatrix} x_0 \\ u_0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$
$$\Rightarrow \begin{vmatrix} s_0 I - A & -b \\ -c^T & -d \end{vmatrix} = 0$$

• If  $s_0$  is not a pole we find

$$\Rightarrow 0 = |s_0 I - A| |d + c^T (s_0 I - A)^{-1} b| =$$
$$a(s_0) P(s_0) \Rightarrow P(s_0) = 0$$

• i.e.  $s_0$  is a zero of the TF

$$\Rightarrow \frac{b(s_0)}{a(s_0)} = 0 \Rightarrow b(s_0) = 0 \text{ since } a(s_0) \neq 0$$

and so a zero of TF numerator.

• But again the SS analysis associates a null direction  $[x_0, u_0]^T$  with the zero.

We say  $s_0$  is a zero with direction  $[x_0, u_0]^T$

If we stimulate the SS system with a signal  $e^{s_0 t}$  but wrong direction we do not get a null output.

## Slide 5

## Direction of the zero (SISO)

To find  $[x_0, u_0]^T$  rewrite defining equation as

$$s_0 \begin{pmatrix} I & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_0 \\ u_0 \end{pmatrix} = \begin{pmatrix} A & b \\ c^T & d \end{pmatrix} \begin{pmatrix} x_0 \\ u_0 \end{pmatrix}$$

i.e.  $[x_0, u_0]^T$  solves a generalised eigenvalue problem and can be computed using an appropriate numerical algorithm.

- SS equations in general can be written

$$\begin{aligned} R(s) \begin{pmatrix} \bar{x} \\ \bar{u} \end{pmatrix} &= \begin{pmatrix} 0 \\ \bar{y} \end{pmatrix} \\ R(s) &= \begin{pmatrix} sI - A & -b \\ -c^T & -d \end{pmatrix} \\ &= \text{Rosenbrock matrix} \\ &= \text{SS system matrix} \end{aligned}$$

- Thus zero direction solves;  $R(s_0)(x_0, u_0)^T = 0$ ; so is a null vector of  $R(s_0)$ .

## Slide 6

## Example 10.1

Find zeros and directions for

$$A_c = \begin{pmatrix} -7 & -12 \\ 1 & 0 \end{pmatrix}, b_c = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$
$$c_c^T = \begin{pmatrix} 1 & 2 \end{pmatrix}, d_c = 0$$

•*TF* view

$$\begin{aligned} P(s) &= c_c^T (sI - A_c)^{-1} b_c = \frac{b(s)}{a(s)} \\ &= (1, 2) \begin{pmatrix} s+7 & 12 \\ -1 & s \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= (1, 2) \frac{1}{s(s+7)+12} \begin{pmatrix} s & -12 \\ 1 & s+7 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= (1, 2) \begin{pmatrix} s \\ 1 \end{pmatrix} \frac{1}{s^2 + 7s + 12} \\ &= \frac{s+2}{s^2 + 7s + 12} = \frac{s+2}{(s+3)(s+4)} \end{aligned}$$

$\Rightarrow$  zero at  $-2$ ; poles at  $-3, -4$

**Slide 7**

ex10a

sys\_10a

a =

|    |    |     |
|----|----|-----|
|    | x1 | x2  |
| x1 | -7 | -12 |
| x2 | 1  | 0   |

b =

|    |    |
|----|----|
|    | u1 |
| x1 | 1  |
| x2 | 0  |

c =

|    |    |    |
|----|----|----|
|    | x1 | x2 |
| y1 | 1  | 2  |

d =

|    |    |
|----|----|
|    | u1 |
| y1 | 0  |

Continuous-time model.

[a,b,c,d]=ssdata(sys\_10a);

[v,d]=eig(a)

v =

|         |         |
|---------|---------|
| -0.9701 | 0.9487  |
| 0.2425  | -0.3162 |

d =

|    |    |
|----|----|
| -4 | 0  |
| 0  | -3 |

help mvzero

```
=====
COMMAND : [z,dir,G]=mvzero(sys,i)
ACTION  : Gets multivariable zeros

INPUTS  : sys = SS system
          i = 1/2 = direct form/indirect form

OUTPUTS : zeros, directions
=====
```

mvzero(sys\_10a,1);

a =

|   |    |     |
|---|----|-----|
|   | -7 | -12 |
| 1 |    | 0   |

b =

|   |
|---|
| 1 |
| 0 |

c =

|   |   |
|---|---|
| 1 | 2 |
|---|---|

d =

|   |
|---|
| 0 |
|---|

Direct Form

Zeroes

|     |
|-----|
| Inf |
| Inf |
| -2  |

Directions

|        |        |         |
|--------|--------|---------|
| 0      | 0.0000 | -0.6667 |
| 0      | 0      | 0.3333  |
| 1.0000 | 1.0000 | 0.6667  |

mvzero(sys\_10a,2);

a =

|   |    |     |
|---|----|-----|
|   | -7 | -12 |
| 1 |    | 0   |

b =

|   |
|---|
| 1 |
| 0 |

c =

|   |   |
|---|---|
| 1 | 2 |
|---|---|

d =

|   |
|---|
| 0 |
|---|

Indirect Form

Zeroes

|         |
|---------|
| 0       |
| 0       |
| -0.5000 |

Directions

|        |         |         |
|--------|---------|---------|
| 0      | 0.0000  | -0.6667 |
| 0      | 0       | 0.3333  |
| 1.0000 | -1.0000 | 0.6667  |



## POLE and ZERO (SISO)

What happens when  $s_0$  is a pole? Let  $q_0$  be the corresponding left evec of  $A$

$$\begin{aligned}\Rightarrow 0 &= s_0 x_0 - A x_0 - b u_0 \\ \Rightarrow 0 &= s_0 q_0^T x_0 - q_0^T A x_0 - q_0^T b u_0 \\ &= s_0 q_0^T x_0 - s_0 q_0^T x_0 - q_0^T b u_0 \\ &= -q_0^T b u_0\end{aligned}$$

$\Rightarrow$  either  $q_0^T b = 0$  i.e. mode is uncontrollable or  $u_0 = 0$ . But  $u_0 = 0$

$$\Rightarrow s_0 x_0 - A x_0 = 0, \text{ \& } c^T x_0 = 0$$

$$\Rightarrow x_0 = p_0 = \text{right evec}$$

$$\Rightarrow c^T p_0 = 0$$

$\Rightarrow$  unobservable mode

• i.e. as expected

a pole-zero cancellation  $\Leftrightarrow$  either lack of controllability or lack of observability.

## SISO Example 10.2

Consider the SISO system with TF

$$P(s) = \frac{s+1}{s(s-1)} = \frac{s+1}{s^2-s} = \frac{b_1s+b_2}{s^2+a_1s+a_2}$$

$$\Leftrightarrow \text{SS: } A_c = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, b = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, c = [1, 1]$$

Clearly  $s_0 = -1$  is a zero.

To find  $x_0$ ,  $u_0$  solve

$$\begin{pmatrix} s_0 x_0 \\ 0 \end{pmatrix} = \begin{pmatrix} A & b \\ c^T & d \end{pmatrix} \begin{pmatrix} x_0 \\ u_0 \end{pmatrix}$$
$$\Rightarrow \begin{pmatrix} -x_{0,1} \\ -x_{0,2} \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_{0,1} \\ x_{0,2} \\ u_0 \end{pmatrix}$$

## Example 10.2 Continued

$$\Rightarrow -x_{0,1} = x_{0,1} + u_0$$

$$-x_{0,2} = x_{0,1}$$

$$0 = x_{0,1} + x_{0,2}$$

$$\Rightarrow x_{0,1} = \frac{u_0}{2} \Rightarrow x_{0,2} = -\frac{u_0}{2}$$

$\Rightarrow u_0$  is a scale factor; set  $u_0 = 2$

$$\Rightarrow (x_{0,1}, x_{0,2}) = (1, -1)$$

$$\Rightarrow u(t) = \mathcal{H}(t)2e^{-t}$$

$$\Rightarrow \dot{x}_1 = x_1 + u$$

$$\Rightarrow x_1 = e^t(-1) + \int_0^t e^{t-\tau} 2e^{-\tau} d\tau$$

$$= -e^t + e^t[-e^{-2\tau}]_0^t = -e^{-t}$$

$$\Rightarrow \dot{x}_2 = x_1 \Rightarrow x_2 = e^{-t}$$

$$\Rightarrow y = e^{-t} - e^{-t} = 0$$

## MIMO POLES

Again look for response of the form  $x = e^{st}x_0$  to null forcing  $\Rightarrow$  get same result as in SISO case

$s_p$  is an eigenvalue of  $A$

i.e.  $|s_p I - A| = 0$ ;

$x_0$  is a right evec of  $A$ .

So  $s_p$  is called a pole with direction  $x_0$ .

## MIMO ZEROES

- $SS$  model

$$\dot{x} = Ax + B_{n \times m}u, y = C_{p \times n}^T x$$

Again look for response to  $u = u_0 e^{s_0 t}$  of the form  $x = x_0 e^{s_0 t}, y = 0$

- We find as before  $s_0, x_0, u_0$  satisfy

$$R(s_0) \begin{pmatrix} x_0 \\ u_0 \end{pmatrix} = 0$$

$$R(s) = \begin{pmatrix} sI - A_{n \times n} & -B_{n \times m} \\ -C_{p \times n}^T & -D_{p \times m} \end{pmatrix}$$

## MIMO ZEROES Continued

- If  $p = m$  so  $R(s)$  is square and if  $s_0$  is not a pole we have

$$\begin{aligned} 0 &= |R(s_0)| \\ &= |s_0 I - A| |D + C^T (s_0 I - A)^{-1} B| \\ &= a(s_0) |P(s_0)| \end{aligned}$$

So a zero solves  $|P(s_0)| = 0$

- Because of the null output property these MIMO zeroes are called *transmission blocking zeroes*.
- When  $p \neq m$  must solve the general eigenvalue problem.

$$s_0 \begin{pmatrix} I & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_0 \\ u_0 \end{pmatrix} = \begin{pmatrix} A & B \\ C^T & D \end{pmatrix} \begin{pmatrix} x_0 \\ u_0 \end{pmatrix}$$

sys\_ex10c

a =

|    | x1 | x2 | x3 |
|----|----|----|----|
| x1 | -2 | 0  | 0  |
| x2 | 0  | -3 | 0  |
| x3 | 0  | 0  | -4 |

b =

|    | u1 | u2 |
|----|----|----|
| x1 | -1 | 0  |
| x2 | -1 | 0  |
| x3 | 0  | 1  |

c =

|    | x1 | x2 | x3 |
|----|----|----|----|
| y1 | 1  | 0  | 0  |
| y2 | 0  | 1  | 1  |

d =

|    | u1 | u2 |
|----|----|----|
| y1 | 1  | 0  |
| y2 | 1  | 0  |

Continuous-time model.

Direct Form of Generalised Eigenvalue Problem

mvzero(sys\_ex10c,1)

Direct Form

Zeroes

|         |   |      |
|---------|---|------|
| -Inf    | - | NaNi |
| -Inf    | - | NaNi |
| -Inf    | - | NaNi |
| -1.0000 |   |      |
| -3.0000 |   |      |

Directions

|         |         |        |         |         |
|---------|---------|--------|---------|---------|
| 0       | 0       | 0      | 0.4588  | 0.0000  |
| 0       | 0       | 0      | 0.2294  | 0.5774  |
| 0       | 0       | 0.0000 | 0.2294  | -0.5774 |
| -1.0000 | 0       | 0      | -0.4588 | 0.0000  |
| 0       | -1.0000 | 1.0000 | 0.6882  | -0.5774 |

Indirect Form of Generalised Eigenvalue Problem

[z,dir]=mvzero(sys\_ex10c,2);

Indirect Form

Zeroes

|         |
|---------|
| -1.0000 |
| -0.3333 |
| 0.0000  |
| 0.0000  |
| 0.0000  |

Directions

|         |         |         |         |         |
|---------|---------|---------|---------|---------|
| 0.4588  | 0.0000  | -0.0000 | 0.0000  | 0.0000  |
| 0.2294  | -0.5774 | -0.0000 | -0.0000 | 0.0000  |
| 0.2294  | 0.5774  | -0.0000 | 0.0000  | -0.0000 |
| -0.4588 | -0.0000 | 0.0000  | -0.0000 | 0.9199  |
| 0.6882  | 0.5774  | -1.0000 | 0.0000  | 0.3922  |

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10.15

```
function [z,dir,G]=mvzero(sys,i) % copyright V Solo 2001
```

```
%=====
%  COMMAND   :   [z,dir,G]=mvzero(sys,i)
%  ACTION    :   Gets multivariable zeros
%
%  INPUTS    :   sys = SS system
%                  i = 1/2 = direct form/indirect form
%
%  OUTPUTS   :   zeros, directions
%=====
disp(' ');
[a,b,c,d]=ssdata(sys);
dim=size(a);
n=dim(1);
G=[[a' c']' [b' d']'];
ai=eye(n);
oc=zeros(size(c));
ob=zeros(size(b));
od=zeros(size(d));
M=[[ai oc']' [ob' od']'];
%
if i==1
    [V,D]=eig(G,M);
    text='Direct Form';
    disp(text);
else
    [V,D]=eig(M,G);
    text='Indirect Form';
    disp(text);
end
z=real(diag(D));
dir=real(V);
%=====
disp(' ');
%
text='Zeroes';
disp(text);
disp(z);
%
text='Directions';
disp(text);
disp(dir);
```



## MIMO ZEROES Continued II

- If  $s_0$  is a pole let  $q_0$  be the corresponding left evec

$$\begin{aligned} s_0 x_0 - A x_0 - B u_0 &= 0 \\ \Rightarrow s_0 q_0^T x_0 - q_0^T A x_0 - q_0^T B u_0 &= 0 \\ \Rightarrow s_0 q_0^T x_0 - s_0 q_0^T x_0 - q_0^T B u_0 &= 0 \\ \Rightarrow q_0^T B u_0 &= 0 \\ &= (B^T q_0)^T u_0 \end{aligned}$$

- But situation is no longer simple as for SISO; cannot conclude either  $q_0^T B = 0$  or  $u_0 = 0$

Conclude that can have pole and zero in the same place without cancellation!

- To get cancellation need directions to align.
- If it happens that  $u_0 = 0$ , then  $s_0 x_0 = A x_0 \Rightarrow x_0 = p_0 =$  right evec and  $C^T x_0 = 0 \Rightarrow c^T p_0 = 0 \Rightarrow$  an unobservable mode.
- If it happens that  $q_0^T B = 0$ , then we have an uncontrollable mode.

## Example 10d Continued

Continuing example 10c, we look at the TF view

$$\begin{aligned} P(s) &= C^T (sI - A)^{-1} B \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{s+2} & 0 & 0 \\ 0 & \frac{1}{s+3} & 0 \\ 0 & 0 & \frac{1}{s+4} \end{pmatrix} \begin{pmatrix} -1 & 0 \\ -1 & 0 \\ 0 & 1 \end{pmatrix} \\ &\quad + \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{s+2} & 0 & 0 \\ 0 & \frac{1}{s+3} & \frac{1}{s+4} \end{pmatrix} \begin{pmatrix} -1 & 0 \\ -1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} -\frac{1}{s+2} & 0 \\ -\frac{1}{s+3} & \frac{1}{s+4} \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} \frac{s+1}{s+2} & 0 \\ \frac{s+2}{s+3} & \frac{1}{s+4} \end{pmatrix} \end{aligned}$$

## Example 10d Continued

To find transmission zeroes not coincident with poles we can solve

$$0 = |P(s)| = \frac{s + 1}{(s + 2)(s + 4)}$$

$\Rightarrow$  zero at  $-1$ ; poles at  $-2, -4$

c.f. SS view which shows;

zeroes at  $-1, -3$  ; poles at  $-2, -3, -4$

i.e. (as expected) TF view misses the coincident pole, zero pair.

## MIMO ZEROES Continued III

Zeros computed from a minimal SS realization are transmission zeroes.

If we don't use a minimal realization we may get additional zeroes called invariant zeroes.

Invariant zeroes consist of input decoupling zeroes and output decoupling zeroes.

Input decoupling zeroes cancel poles with uncontrollable modes

Output decoupling zeroes cancel poles with unobservable modes

Transmission zeroes + invariant zeroes = system zeroes

Should use minimal realization to avoid these other zeroes.

## POLES & ZEROES: Transfer function view

Convert  $P(s)$  to SMF

$$P(s) = U(s)_{p \times p} M(s)_{p \times m} V(s)_{m \times m}$$
$$M(s) = \left[ \begin{array}{c|c} \text{diag}(\frac{\epsilon_i(s)}{\psi_i(s)}) & 0 \\ \hline 0 & 0 \end{array} \right] \begin{array}{l} r \\ p - r \end{array}$$

$r \qquad m - r$

Define

zeroes are roots of  $\epsilon_i(s), i = 1, \dots, r$

poles are roots of  $\psi_i(s), i = 1, \dots, r$

### Example 10e

$$P(s) = \begin{pmatrix} \frac{s+1}{s+2} & 0 \\ \frac{s+2}{s+3} & \frac{1}{s+4} \end{pmatrix} = \frac{N(s)}{d(s)}$$

$$\Rightarrow N(s) = \begin{pmatrix} (s+1)(s+3)(s+4) & 0 \\ (s+2)^2(s+4) & (s+2)(s+3) \end{pmatrix}$$

$$\Rightarrow d(s) = (s+2)(s+3)(s+4)$$

$$\Delta_0(s) = 1$$

$$\Delta_1(s) = \gcd(\text{all } 1 \times 1 \text{ elements}) = 1$$

$$\begin{aligned} \Delta_2(s) &= \gcd(\text{all } 2 \times 2 \text{ minors}) \\ &= (s+2)(s+3)^2(s+1)(s+4) \end{aligned}$$

## Example 10e Continued

$$\Rightarrow \lambda_1(s) = \frac{\Delta_1}{\Delta_0} = 1$$

$$\lambda_2(s) = \frac{\Delta_2}{\Delta_1} = (s+2)(s+3)^2(s+1)(s+4)$$

$$\Rightarrow \Lambda(s) = \begin{bmatrix} 1 & 0 \\ 0 & (s+1)(s+2)(s+3)^2(s+4) \end{bmatrix}$$

$$\Rightarrow M(s) = \begin{bmatrix} \frac{1}{(s+2)(s+3)(s+4)} & 0 \\ 0 & (s+1)(s+3) \end{bmatrix}$$

$\Rightarrow$ zeros at  $-1, -3$ ;  $\Rightarrow$ poles at  $-2, -3, -4$

which agrees with the SS computation.

We prove equivalence below

## POLES & ZEROS - MFD view

Let  $P(s)$  have irreducible MFDs

$$P(s) = N_R(s)D_R^{-1}(s) = D_L^{-1}(s)N_L(s)$$

- $s_p$  is a pole iff

$D(s)$  loses rank at  $s_p$

- Since  $a(s) = \det D(s)$

$\Rightarrow s_p$  are roots of  $a(s)$

- $s_0$  is a transmission zero of  $P(s)$

iff  $N(s)$  loses rank at  $s_0$



## Irreducible MFDs

If  $N_1 D_1^{-1}, N_2 D_2^{-1}$  are two irreducible right MFDs of  $P(s)$  then they have a unimodular divisor

$$N_1 = N_2 U, D_1 = D_2 U$$

Proof. By the gcd construction we have

$$U \begin{pmatrix} D_1 \\ N_1 \end{pmatrix} = \begin{pmatrix} R \\ 0 \end{pmatrix}$$

where  $U, R$  are both unimodular

But  $N_1 D_1^{-1} = N_2 D_2^{-1} \Rightarrow N_1 = N_2 D_2^{-1} D_1$

## Irreducible MFDs Continued

So rewrite as

$$\begin{aligned}\begin{pmatrix} R \\ 0 \end{pmatrix} &= U \begin{pmatrix} D_2 D_2^{-1} D_1 \\ N_2 D_2^{-1} D_1 \end{pmatrix} = U \begin{pmatrix} D_2 \\ N_2 \end{pmatrix} D_2^{-1} D_1 \\ \Rightarrow U \begin{pmatrix} D_2 \\ N_2 \end{pmatrix} &= \begin{pmatrix} R & 0 \\ 0 & I \end{pmatrix} \begin{pmatrix} D_1^{-1} D_2 \\ 0 \end{pmatrix} \\ \Rightarrow \begin{pmatrix} R^{-1} & 0 \\ 0 & I \end{pmatrix} U \begin{pmatrix} D_2 \\ N_2 \end{pmatrix} &= \begin{pmatrix} D_1^{-1} D_2 \\ 0 \end{pmatrix}\end{aligned}$$

But  $R^{-1}$  is a polynomial matrix so we deduce  $D_1^{-1} D_2$  is a polynomial matrix. Interchange 1, 2 in above argument to find  $D_2^{-1} D_1$  is a polynomial matrix  $\Rightarrow D_2^{-1} D_1$  is unimodular.

$\Rightarrow N_1 = N_2(D_2^{-1} D_1), D_1 = D_2(D_2^{-1} D_1)$  as required.

## SMF revisited

From the SMF we can construct left or right irreducible MFDs.

Firstly write

$$\begin{aligned}M(s) &= E(s)\Psi_R^{-1}(s) = \Psi_L^{-1}(s)E(s) \\ E(s)_{p \times m} &= \begin{bmatrix} \text{diag}(\epsilon_i(s) & 0 \\ 0 & 0 \end{bmatrix} \\ \Psi_R(s) &= \begin{bmatrix} \text{diag}(\psi_i(s) & 0 \\ 0 & I_{m-r} \end{bmatrix} \\ \Psi_L(s) &= \begin{bmatrix} \text{diag}(\psi_i(s) & 0 \\ 0 & I_{p-r} \end{bmatrix}\end{aligned}$$

## SMF revisited Continued

Then

$$\begin{aligned}P(s) &= U(s)M(s)V(s) \\&= [U(s)E(s)][\Psi_R^{-1}(s)V(s)] \\&= N_R(s)D_R^{-1}(s) \\D_R(s) &= V^{-1}(s)\Psi_R(s) = \text{polynomial matrix}\end{aligned}$$

also

$$\begin{aligned}P(s) &= U(s)\Psi_L^{-1}(s)E(s)V(s) \\&= D_L^{-1}(s)N_L(s) \\D_L(s) &= \Psi_L(s)U^{-1}(s) = \text{polynomial matrix}\end{aligned}$$

And these MFDs are irreducible (otherwise  $\epsilon_i(s), \psi_i(s)$  would have uncanceled common factors)

## SMF and MFDs

• Let  $N(s)D^{-1}(s)$  be any other irreducible right MFD of  $P(s)$ . By the MFD lemma there is a unimodular matrix  $\overline{U}(s)$  with

$$\begin{aligned} N(s) &= N_R(s)\overline{U}(s) \\ &= U(s)E(s)\overline{U}(s) \end{aligned}$$

and (putting  $V_-(s) = V^{-1}(s)$  = unimodular matrix)

$$\begin{aligned} D(s) &= D_R(s)\overline{U}(s) \\ &= V_-(s)\Psi_R(s)\overline{U}(s) \\ &= \text{Smith form of } D(s) \end{aligned}$$

• Thus

$$\begin{aligned} |D(s)| &= |V_-(s)| |\Psi_R(s)| |\overline{U}(s)| \\ &= \text{const} \times \prod_1^r \psi_i(s) \end{aligned}$$

So roots of  $|D(s)| = 0$  are roots of  $\psi_i(s)$

## SMF and MFDs Continued

$$\begin{aligned}\Rightarrow \text{MFD poles} &\equiv \text{SMF poles} \\ &\equiv \text{Kalman SS poles} \\ &\equiv \text{poles of any minimal SS realization}\end{aligned}$$

### • Continuing

$$N(s_0) = U(s_0)E(s_0)\overline{U}(s_0)$$

Now  $|U(s_0)| = \text{const}$ ,  $|\overline{U}(s_0)| = \text{const}$ . So

$U(s_0), \overline{U}(s_0)$  are nonsingular. So  $N(s_0)$  loses rank  
iff  $E(s_0)$  loses rank i.e. iff  $\epsilon_i(s_0) = 0$  for some  $\epsilon_i(s)$

$$\begin{aligned}\Rightarrow \text{MFD zeros} &\equiv \text{SMF zeros} \\ &\equiv \text{Kalman SS zeros} \\ &\equiv \text{zeros of any minimal SS realization}\end{aligned}$$

## MIMO Zeros and Transfer Function

In the rectangular case  $p \neq m$  can characterise the input direction  $u_0$  from the transfer function.

We have

$$\begin{aligned}(s_0 I - A)x_0 &= Bu_0 \\ C^T x_0 + Du_0 &= 0\end{aligned}$$

So long as  $s_0$  is not a pole we get

$$\begin{aligned}x_0 &= (s_0 I - A)^{-1} Bu_0 \\ \Rightarrow 0 &= C^T x_0 + Du_0 \\ &= C^T (s_0 I - A)^{-1} Bu_0 + Du_0 \\ &= P(s_0)u_0\end{aligned}$$

So a transmission zero that is not a pole causes a loss of rank in  $P(s)$  and  $u_0$  is the corresponding null vector.

N.B. MFD view:  $P(s) = D_L^{-1}(s)N_L(s)$

(irreducible)  $\Rightarrow$  So long as  $s_0$  is not a pole  $D_L(s_0)$  has full rank so  $P(s_0)u_0 = 0 \leftrightarrow N_L(s_0)u_0 = 0$ . So can determine  $u_0$  also from the MFD.